

PQC Scanner - Supported Formats & Capabilities

Dedicated Reference for Post-Quantum Cryptography Readiness Engine

1. PQC Scanner - Supported Extensions & File Types

The PQC Scanner parses cryptographic configurations, certificates, keystores, and application manifests across 21 supported file extensions:

File Extensions	PQC Audit Scope & Target Configs
.conf / .cnf / .config / .ini / .cfg	Web Servers (NGINX, Apache, HAProxy), DBs (MySQL, MariaDB), OpenSSL
.yaml / .yml	Kubernetes Secrets (kubernetes.io/tls), Ingress manifests, Docker Compose
.properties / .toml	Spring Boot (application.properties), Java app configs, Rust/Rustls TOML
.pem / .crt / .cer	X.509 Certificate Chains, TLS/SSL Certificates, CA Bundles
.key	Private Key exports (RSA, ECC, Ed25519, Curve25519, DSA)
.p12 / .pfx / .jks	PKCS#12 Keystores, Java KeyStores, IIS SSL Certificate Bundles
.json / .xml	AWS KMS/IAM policies, Azure Key Vault exports, GCP Cloud KMS
.txt / .pdf / .html	Extracted evidence text, SSL/TLS scan logs, cryptographic audit documents

2. PQC Scanner - Target Software, Services & Infrastructure

- * **Web Servers & Reverse Proxies:** NGINX, Apache HTTPD, HAProxy, Envoy Proxy, IIS, Traefik
- * **Database Engines:** MySQL (9.x/8.x), MariaDB, PostgreSQL, MS SQL Server, Oracle DB, MongoDB
- * **Cloud Services & Containers:** Kubernetes Secrets (kubernetes.io/tls), Ingress, Docker Compose, AWS KMS, Azure Key Vault, GCP KMS
- * **Enterprise Applications:** Spring Boot (application.properties/yml), PKCS#12, Java KeyStores (.jks)
- * **VPN & Network Security:** StrongSwan/Libreswan IPSec (IKEv2 modp DH groups), OpenVPN, WireGuard, Palo Alto, Fortinet
- * **Infrastructure & SSH:** OpenSSH (sshd_config - KexAlgorithms, Ciphers, HostKeyAlgorithms)

3. Cryptographic Classifications & NIST PQC Standards

- * **QUANTUM-VULNERABLE:** RSA (512-4096), ECC (P-256/P-384/P-521, secp256k1), Curve25519/X25519, Ed25519, ECDSA, ECDH, Diffie-Hellman (DH Groups 1-24, modp1024-modp8192), DSA, DB TLS without PQC KEM
- * **CLASSICALLY-WEAK:** MD5, SHA-1, 3DES, DES, RC4, SSLv2, SSLv3, TLS 1.0, TLS 1.1, CBC-mode ciphers
- * **QUANTUM-SAFE (NIST):** ML-KEM (FIPS 203 / Kyber), ML-DSA (FIPS 204 / Dilithium), SLH-DSA (FIPS 205 / SPHINCS+), Falcon (NIST IR 8413), AES-256-GCM, SHA-384, SHA-512, ChaCha20-Poly1305

4. PQC Cryptographic Inventories & Risk Metrics (Zero CVSS)

* CBOM:	Cryptographic Bill of Materials - inventory of all cryptographic assets, keys, certificates, and cipher suites.
* QBOM:	Quantum Bill of Materials - CBOM annotated with quantum readiness status (VULNERABLE / WEAK / SAFE).
* PQC Risk Bands:	P1 Critical, P2 High, P3 Medium, P4 Low - pure PQC risk metrics, zero CVSS scores.
* Quantum Risk Score:	Deterministic risk score (0 to 100) based on algorithm vulnerability, key size, and exposure context.

5. PQC Migration Roadmap & OEM Vendor Matrix

- Asset Prioritization Tiers:	4-Tier migration classification (Tier 1: Immediate HNDL Risk -> Tier 4: Low Impact).
- OEM Readiness Status:	Vendor readiness tracking: Upgrade Required, In Development, PQC Ready, Review Required.
- Migration Dependency Chain:	Dependency chain mapping between software components, OS libraries, and HSM/KMS hardware.
- Actionable Remediation:	Specific NGINX, OpenSSL, Java, and SSH configuration steps with NIST FIPS 203/204 references.

6. Dedicated PQC Report Export Formats

- PQC PDF Report (_export_pqc_pdf):	Dedicated PQC PDF featuring CBOM, QBOM, P1-P4 Risk Bands, OEM Matrix, and NIST FIPS 203/204/205 standards. Zero VAPT headers or CVSS scores.
- PQC Word Document (_export_pqc_docx):	Fully styled Microsoft Word (.docx) report with PQC executive tables, risk callouts, and migration roadmaps.
- PQC Excel Worksheet (export_excel_report):	Complete tabular export containing CBOM/QBOM asset registers, algorithm findings, and remediation task lists.
- PQC Raw JSON Export (export_json_report):	Machine-readable JSON data export for SIEM, GRC, or security dashboard integration.